

CSCE 5380 : DATA MINING

PROJECT 02 MILE STONE 02 REPORT

A. Task 1a: Description of Preprocessing Operations:

The important step in preparing the dataset for K-Means clustering algorithm is make that every data feature is of numeric type as its based on distances between points to cluster the data. From summary I found two variables the CALC and NObeyesdad are the two character features, wo we need to deal with and make sure all the features to be scaled. The first step I have done is to check for missing values which I found that there are none. This is important as null values may impact the distance calculations and may lead to wrong analysis and results. Next I dealt to check and removing the duplicates by using the unique function. This will help us in avoiding the bias.

Moving on to the second step, I have done is to get the numerical values for the ordinal categorical features like calc which is associated with alcohol consumption from the description file has four unique values from No to Always, so we assign it to be scale of 1 to 4. Similarly, we assign the values for NObeyesdad feature for the Insufficient weight to obese on a scale of 1 to 4 again. This method ensures the ranking and order of the numbers maintained correctly in the features, Next I ensured once again that all the features are numeric. For the final step I made sure that I scaled everything as the features are on different scales which effect the Euclidean distance, as features with larger values will dominate the clustering process in distance measurements making others not valuable. This standardization ensures that all features are considered equally. Finally saved the data into dataframe that s ready for the clustering procedure in task2.

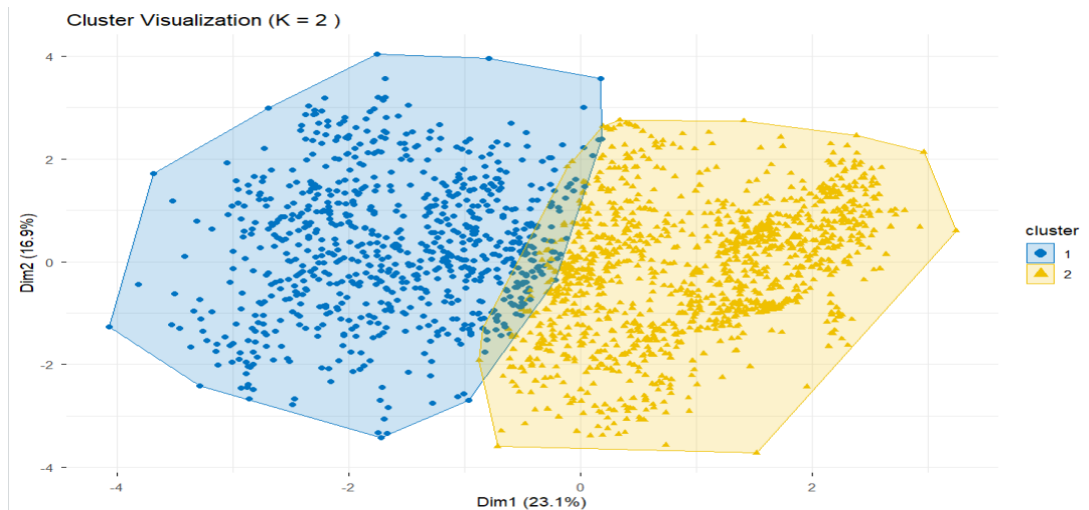
B. Task 2a: K means with K=2:

The K means with k=2 is successfully implemented and two cluster sizes are 804, 1282.

The centroid values from the cluster summary being as below table:

Cluster	Count	Age	Height	Weight	FCVC	NCP	CH2O	FAF	TUE	CALC	NObeyesdad
1	804	20.7	1.67	61.6	2.33	2.75	1.87	1.19	0.817	1.6	2.03
2	1282	26.6	1.73	103	2.48	2.67	2.09	0.9	0.567	1.81	3.73

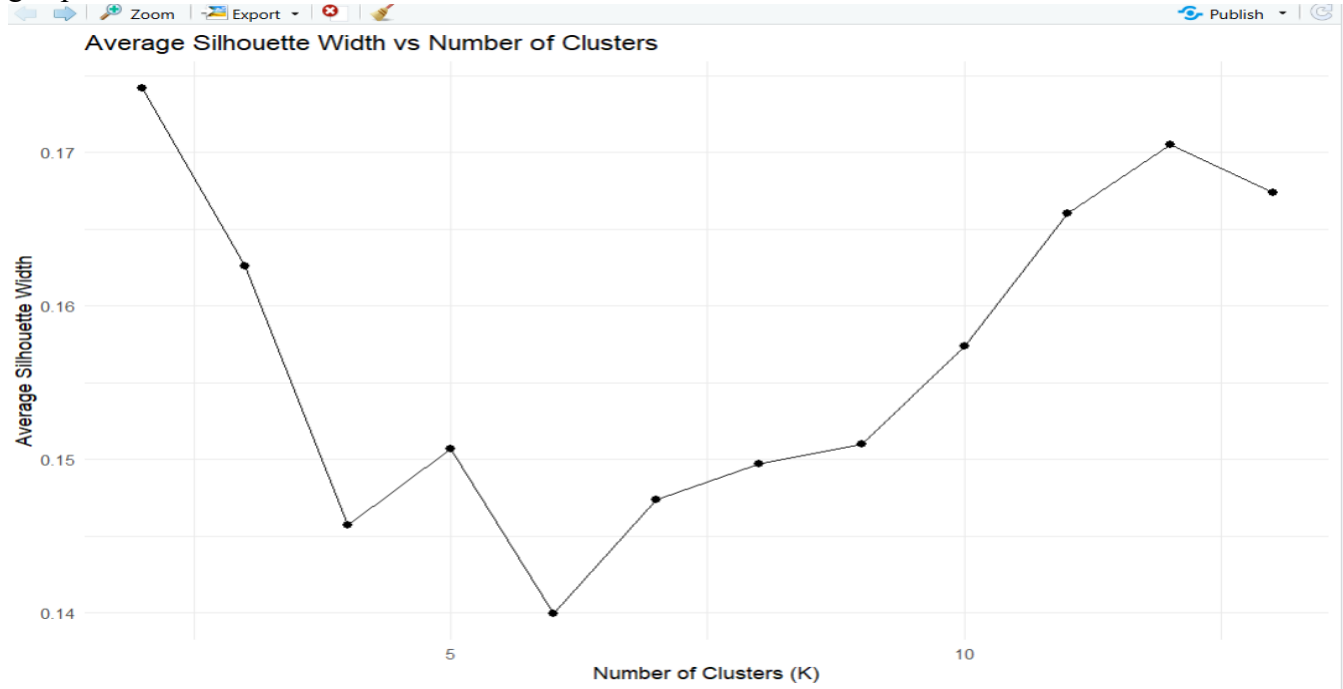
The plot of K=2 is



Here dimensions 1 and 2 are from PCA of factoviz visualization function fviz() creates them (Not done manually) for easy visualization as there is many features which can't be plot on 3d system we know.

C. Task 2b: Finding the optimal K using the silhouette Analysis:

We performed the silhouette Analysis for $k=2$ to 13 and plotted the K vs score in a graph. The $K=2$, has the highest silhouette score meaning the cluster width is high for $k=2$, and optimal value should be $K=2$, Although $k=12$ almost reached but it has surpassed and $k=2$ score and , 12 will too complex for us to grasp.



D. Task 3a: Plot of feature contribution to each cluster:

From the task 2b, we got the optimal cluster configuration to be $K=2$. And the plotted graph of feature contribution to each cluster of the two is derived as below:



E. Task 3b: Interpretation of Feature Contributions:

From the above feature contribution plot we see how each factor varies across the two optimal clusters derived, based on their standardized mean values. Positive value indicates that the feature values in that cluster is higher than the overall average of all the values, while negative means lower than the overall average value. From the plot we can see that cluster 1 is mainly identified to have lower negative values for features like Weight, Age, Height, and Nobeyesdad. Especially The Weight and NOBeyesdad showing the strong negative relation, indicating the person categorized in this category will have less weight and lower obesity levels, but however they positive relation to physical and technological usage showing that they might have a active lifestyle.

But in case of the cluster 2, we can see that the Weight, Age, Height and Nobeyesdad have higher and positive values indicates opposite of cluster 1, showing the persons in this category have more weight and obese conditions. The age being higher than cluster 1, suggests that this group people are more aged. This cluster also have higher values for alcohol and water intake while showing lower physical activity(FAF) which suggests that they less active lifestyle. At last, we can conclude, the main features that helped distinguish between the two clusters are Weight, Nobeyesdad, Age and lifestyle related variables like FAF, CALC, TUE. The cluster 1 represents the datapoints with lower obese and relatively more active lifestyles whereas cluster 2 represents individuals with higher obesity levels and less active lifestyles. From these we can both physical characteristics and lifestyle factors play a major role in defining the clusters.

Observation Report for CSCE5380 on Project02**F. Task 4: Reflective Report on Association Rule Mining and Clustering:**

In this project 2, I have worked on Association Rule Mining (ARM) for the milestone1 and K Means clustering for the milestone2. The two approaches gave me a idea on how each works on the real-life datasets along with the difference of the knowledge that they each provide, their computational behaviour and their limitations. They also made me understand that how different methods provide different type of knowledge on the same type of data.

The main difference that I observed is ARM provides information on the relationship among the variables which is expressed in form rules, like identifying the behaviour or what the variable value is goanna be on the values of other variables. These rules are like if then formats and based on the measures such as Support, Confidence and Lift which makes the rules easy to understand and easily apply in real-life situations. For example, In the vote analysis, it helps governments how to tailor the bills which can be pushed together to get voted yes. Also helps us determine whether the bill will be voted yes based on which type of bills are votes yes before. (like market basket analysis in the lectures). Meanwhile, in case of clustering we will not get the rules, but we group similar datapoints together to get a small number of clusters according to the features. In this project we used clustered the obese data set into two clusters based on similar physical and lifestyle factors. Here the info is more like a finding as it shows internal form of the data, not direct relationships like in ARM. Thy type can also be applied for customer segmentation, health profiling, or pattern recognition. In our project the division helps determines with groups with obesity problems which can be used for targeted intervention by the companies in health care, cosmetic hospitals.

While coming to the difference in computation power, both methods are very different. ARM uses algorithms such as Apriori are generally computationally more expensive than K means clustering. As the ARM process involves the building of possible sets of data and identifying the prevalence of data for derived set. So the size possible data will increase with no. of variables and ARM may become in efficient with the high dimensional datasets. On the other hand, K Means tends to be linear in terms amount of data, clusters as data is pre-processed before dividing into clusters. Thus, it the computational time required is very much less compared to ARM. But to find the optimal value of k, we might run it more than once which adds to the additional computational cost

Next, the two techniques have own weaknesses as I observed in this project. In case of ARM there is a high degree of rule generation and a high probability that most rules generated are going to be useless and meaningless. The findings are also dependent on measures like support and lift which makes them highly sensitive. If we keep the threshold values to small many rules be there and very high important ones may be missed with too little rules. Also, ARM doesn't assume the casual relations between the features. Moving on to K Means also many limitations. Firstly, the correct of K i.e., the optimal number of clusters is very important, if its too small we may miss some patterns, too large then it will be to complex to understand with human brain. In our project we used the silhouette method to find the K, however the scores are not high (nearer to 1). Which means clusters not so strong, but the optimal came out to be K=2. Sometimes it may not give the well-defined clusters, and the initial position of centroids selected is also important and assumptions of clusters are spherical.

Finally, to conclude, that both are complement to each other. ARM gives understanding of relation between the features whereas K-Means clustering is of grouping the similar datapoints. They both provide needed insights into integrating structural patterns with relational insights. The combination of the two approaches helped me to get a more comprehensive and in-depth insight into the dataset, which is why it is essential to employ a variety of methods in data analysis. This project also gives that no single method can provide all the insight needed and one should use the method based on the required knowledge and details. ARM and clustering are useful in data mining, though they have different applications.